LearningOS test fixture - synthetic page one.
Merge sort splits the array in half, sorts each half, then merges.
The recurrence is T(n) = 2T(n/2) + n, which solves to Theta(n log n).
Case 2 of the master theorem applies because f(n) = n matches n^log_b(a).
A stable merge preserves the relative order of equal keys.
This paragraph exists only so the extraction looks healthy to the indexer.